Perovskite switching: Exercises

> Reference: Jennifer Rupp, Daniele Ielmini, Ilia Valov (eds.), Resistive Switching: Oxide Materials, Mechanisms, Devices and Operations, Springer, 2022. DOI: 10.1007/978-3-030-42424-4.

Unless stated otherwise, the numbers are hypothetical teaching assumptions, not measured device specifications.

E01 Area scaling with a parallel path

Assume an interface with $RA = 100\,\Omega\,\mu\mathrm{m}^2$ at fixed state and bias. Calculate its resistance for areas 10 and $40\,\mu\mathrm{m}^2$. Add a separate, area-independent $100\,\Omega$ parallel path and calculate each total. Does near inverse-area scaling uniquely prove a uniform interface mechanism?

E02 Depletion width

For the one-sided depletion approximation, assume uniform fully ionized donors, $\epsilon_r = 100$, $\epsilon_0 = 8.854 \times 10^{-12}\,\mathrm{F/m}$, $\phi = 0.5\,\mathrm{V}$, $q = 1.602 \times 10^{-19}\,\mathrm{C}$ and $N_D = 10^{18}\,\mathrm{cm}^{-3}$. Find $W = \sqrt{2\epsilon\phi/(qN_D)}$ and the width when only N_D is quadrupled. Name two model limitations.

Lin Cing https://github.com/xinhi6716/resistive-memory-self-study

E03 A log–log slope

For $I = aV + bV^2$ at positive bias, let $a = 1\,\mu\mathrm{A/V}$ and $b = 1\,\mu\mathrm{A/V}^2$. Derive $d\ln I/d\ln V$ and compute it at $V = 0.1, 1, 10\,\mathrm{V}$. Does a slope near 2 uniquely establish trap-free SCLC?

E04 Oxygen pressure and charge balance

For ideal dilute, isothermal equilibrium of $O_O^{\times} \rightleftharpoons \frac{1}{2}O_2 + V_O^{\bullet\bullet} + 2e'$, assume the only relevant charge carriers satisfy $n = 2[V_O]$, and the lattice activity is constant. Derive the exponent of n versus p_{O_2}. What happens to n when the pressure rises by a factor of 64? State when the conductivity would follow the same factor.

E05 SCLC assumptions

Starting with $J = qn\mu E$ and $dE/dx = qn/\epsilon$, derive $J = 9\epsilon\mu V^2/(8d^3)$ for a planar trap-free device under suitable ideal boundaries. List the assumptions and give one reason an observed slope of 2 does not establish the mechanism.

Perovskite switching: Exercises (continued)

E06 Read current and reliability

An ideal device is read at fixed $1\,\mathrm{V}$. Its HRS is $1\,\mathrm{M\Omega}$ and LRS is $10\,\mathrm{k\Omega}$. Calculate the read currents. Can these single numbers guarantee a low-error threshold or long retention? Identify relevant distributions or disturbances.

E07 Polarization and trapping

A resistance–voltage sweep has hysteresis. Explain how ferroelectric polarization and trapped or ionic charge can each affect an interface, why hysteresis alone cannot distinguish them, and propose two complementary controlled observations.

E08 Ring isolation

In a laterally conducting film an ideal continuous high-resistance ring spans the film thickness, enclosing an interior that was not written. Explain why an outside-to-inside path is blocked and how a local low-resistance gap can change the connectivity. Does the observation directly establish oxygen concentration or a uniform interior state?

E09 Humidity and chemical evidence

Give a charge-balanced hydration equation involving water, $V_O^{\bullet\bullet}$ and a lattice oxygen site. A particular device switches at intermediate humidity but not at tested dry or saturated conditions. Design a dry → intermediate-humidity → dry comparison with an untreated control to separate reversible environment effects from irreversible aging; can this result be generalized to all devices?

E10 A partially switched bridge

An initially uniform rectangular bridge has length $200\,\mu\mathrm{m}$, width $20\,\mu\mathrm{m}$ and resistance $R_0 = 1\,\mathrm{M\Omega}$. A $10\,\mu\mathrm{m}$-long band changes over half the width to a resistivity 100 times larger; the other half stays unchanged. Treat the two band parts as parallel, the outside segments as series, and ignore contact resistance and current spreading. Find total resistance and explain why it does not uniquely specify the switched geometry. Propose two complementary control experiments to test whether the assumed geometry or contacts explain the observed resistance.

Perovskite switching: Exercises (continued)

E11 Oxygen-vacancy fraction and charge balance

A hypothetical material has $N_O = 5 \times 10^{22}$ oxygen sites per cm^3. The oxygen-site vacancy fraction f rises from 0.001 to 0.004. Assume all created vacancies are doubly positive and fully ionized and that mobile electrons alone compensate the additional charge; all other charged species remain fixed. Find the change in vacancy concentration and electron concentration. Explain why this is not an unconditional prediction for a real correlated oxide.

E12 An interlayer changes the voltage division

For a hypothetical linear series circuit, an oxide resistance is $R_o = 1\,\mathrm{k\Omega}$, and a separate interlayer/contact resistance is $R_s = 9\,\mathrm{k\Omega}$. A total of $1\,\mathrm{V}$ is applied. Find the current and oxide voltage. Repeat if only R_s is reduced to $4\,\mathrm{k\Omega}$. Why can this simplified result not directly predict switching in a nonlinear device?

Lin Cing
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